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# Determination of the tangent stiffness tensor in materials modeling in case of large deformations by calculation of a directed strain perturbation

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**Abstract**

The Finite Element Method in the field of materials modeling is often relying to the tangent stiffness tensor of the constitutive law. This so called Jacobian matrix is required at each time increment and describes the local material behavior. It assigns a stress increment to a strain increment and is of fundamental importance for the numerical determination of the equilibrium state. For increasingly sophisticated material models the tangent stiffness tensor can be derived analytically only with great effort, if at all. Numerical methods like the forward-difference, the central-difference, and the complex-step derivative approximation approach are widely used for its calculation. For each of these methods it is necessary to generate a specific strain perturbation. However, in large strain formulations it is not possible to perturb the strain directly but one can only modify the deformation gradient.

We present our methods to generate a directed strain perturbation for the Green–Lagrange, Euler–Almansi and logarithmic strain measures as a function of the deformation gradient and compare them with other commonly used methods. An increase in accuracy and rate of convergence can be achieved with the proposed procedures.
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*Keywords:* Jacobian matrix; Tangent stiffness tensor; Numerical calculation; Directed strain perturbation; Materials modeling

## 1. Introduction

Commercial finite element software as ABAQUS [1] or MARC [2] is widely used in the field of materials research. They allow the user to incorporate a subroutine describing the material behavior. This subroutine is called at each integration point and provides the local stress response $\boldsymbol{\sigma}$ and the tangent stiffness tensor $\frac{\mathrm{d}\boldsymbol{\sigma}}{\mathrm{d}\boldsymbol{\varepsilon}}$, where $\boldsymbol{\varepsilon}$ is the local strain, to the solver of the finite element software. Thereby the tangent stiffness tensor not only affects the rate of convergence to the state of equilibrium, but also specifies the direction in stress–strain space in which it is sought. Hence, the determined state of equilibrium depends significantly on the correctness of $\frac{\mathrm{d}\boldsymbol{\sigma}}{\mathrm{d}\boldsymbol{\varepsilon}}$. The most direct way is to derive the stiffness analytically as conducted in [3–5]. For sophisticated material models, e.g. multiscale [6] or crystal plasticity

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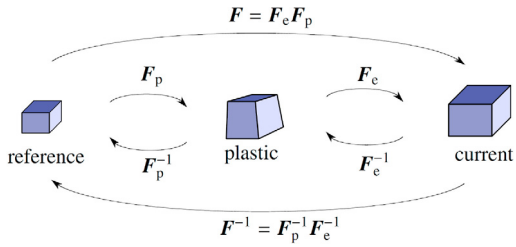


Fig. 1. Multiplicative decomposition of the deformation gradient $\boldsymbol{F}$ into an elastic and a plastic part, and denotation of the different configurations.

models [7], however, it is complex and inflexible to derive an analytical expression. In this context it becomes convenient to conduct the derivation numerically. In order to reduce the computational effort Starman [5] decomposes the tangent stiffness tensor into a part which can be calculated analytically and a part determined numerically. Irrespective of whether you decompose the stiffness tensor or not, you come to the point where you have to perform a numerical approximation. The common approach for the numerical calculation of the tangent stiffness tensor is to conduct a controlled perturbation at the assumed state of equilibrium [8–13]. The general idea for this is outlined in Section 3. In Section 4 of this contribution, we present a survey of both established and novel strategies [10–13]. Since it is not possible to perform a strain perturbation directly in a large strain formulation, one can only modify the deformation gradient. In Section 5 we present a new approach how this deformation gradient can be determined precisely depending on a prescribed strain perturbation. As demonstrated in Section 6 our methods are able to increase the accuracy of the numerical approximation of the tangent stiffness tensor because the approximation works the better the preciser the strain perturbation was determined. Since one key feature which discriminates the different methods is the coordinate system they are based on, we prepend in Section 2 a short recapitulation of the relevant continuum mechanics basics.

## 2. Configurations and strain measures

In continuum mechanics, we distinguish mainly between the reference configuration, generated by the Lagrangian basis vectors and describing the initial, undeformed state, and the current configuration, generated by the Eulerian basis vectors and describing the current, deformed state [14,15], Fig. 1. The fundamental kinematic variable, the deformation gradient $\boldsymbol{F} := \frac{\mathrm{d}\boldsymbol{x}}{\mathrm{d}\boldsymbol{X}}$, transforms an infinitesimal vector $\mathrm{d}\boldsymbol{X}$ within the reference configuration to its new dimension $\mathrm{d}\boldsymbol{x}$ within the current configuration. Since only elastic strains cause stresses, the deformation gradient $\boldsymbol{F}$ is decomposed multiplicatively into the elastic and the plastic deformation gradients $\boldsymbol{F}_\mathrm{e}$ and $\boldsymbol{F}_\mathrm{p}$, so that $\boldsymbol{F} = \boldsymbol{F}_\mathrm{e}\boldsymbol{F}_\mathrm{p}$. Thus we introduced the plastic configuration which describes the state of purely plastic deformation. The elastic Green–Lagrange strain $\boldsymbol{E}^\mathrm{e}$ is defined within the plastic configuration:

$$\boldsymbol{E}^\mathrm{e}(\boldsymbol{F}_\mathrm{e}) := \frac{1}{2}(\boldsymbol{F}_\mathrm{e}^\mathrm{T}\boldsymbol{F}_\mathrm{e} - \boldsymbol{I}), \tag{1}$$

where $\boldsymbol{I}$ is the $3 \times 3$ unit tensor. Within the current configuration we denote the elastic strain $\boldsymbol{\varepsilon}^\mathrm{e}$ and distinguish between the logarithmic ($\boldsymbol{\varepsilon}^\mathrm{e}_\mathrm{log}$) and the Euler–Almansi strain measure ($\boldsymbol{\varepsilon}^\mathrm{e}_\mathrm{EA}$):

$$\boldsymbol{\varepsilon}^\mathrm{e}_\mathrm{log}(\boldsymbol{F}_\mathrm{e}) := \frac{1}{2}\ln(\boldsymbol{F}_\mathrm{e}\boldsymbol{F}_\mathrm{e}^\mathrm{T}), \tag{2}$$

$$\boldsymbol{\varepsilon}^\mathrm{e}_\mathrm{EA}(\boldsymbol{F}_\mathrm{e}) := \frac{1}{2}(\boldsymbol{I} - (\boldsymbol{F}_\mathrm{e}\boldsymbol{F}_\mathrm{e}^\mathrm{T})^{-1}). \tag{3}$$

Performing the double contraction with the elastic tangent stiffness tensor $\mathbb{C}_\mathrm{e}^\mathrm{pla}$ (defined within the plastic configuration) or $\mathbb{C}_\mathrm{e}^\mathrm{cur}$ (defined within the current configuration), we obtain the 2nd Piola–Kirchhoff stress $\boldsymbol{S}$ within the plastic configuration and the Cauchy stress $\boldsymbol{\sigma}$ within the current configuration:

$$\boldsymbol{S} := \mathbb{C}_\mathrm{e}^\mathrm{pla} : \boldsymbol{E}^\mathrm{e}, \tag{4}$$

$$\boldsymbol{\sigma} := \mathbb{C}_\mathrm{e}^\mathrm{cur} : \boldsymbol{\varepsilon}^\mathrm{e}. \tag{5}$$

Using the elastic deformation gradient $\boldsymbol{F}_\mathrm{e}$ we can transform any tensor from the plastic to the current configuration and vice versa [16].

In contrast, the so-called intermediate configuration, which is not explicitly labeled in Fig. 1, involves information from the reference and the current configuration simultaneously. The stress measure within this configuration, the 1st Piola–Kirchhoff stress $\boldsymbol{P}$, can be derived from the 2nd Piola–Kirchhoff stress or the Cauchy stress:

$$\boldsymbol{P} = \det(\boldsymbol{F}_\mathrm{p})\boldsymbol{F}_\mathrm{e}\,\boldsymbol{S}\,\boldsymbol{F}_\mathrm{p}^{-\mathrm{T}}, \tag{6}$$

$$\boldsymbol{P} = \det(\boldsymbol{F})\,\boldsymbol{\sigma}\,\boldsymbol{F}^{-\mathrm{T}}. \tag{7}$$

## 3. Finite difference approximation of the tangent stiffness tensor

Commonly, the elastic tangent stiffness tensors $\mathbb{C}_\mathrm{e}^\mathrm{pla}$ and $\mathbb{C}_\mathrm{e}^\mathrm{cur}$ are defined depending on Young's modulus $E$ and Poisson's ratio $\nu$ (Hooke's law). However, the elasto-plastic tangent stiffness tensor within the reference, current or intermediate configuration includes the elastic as well as the plastic material behavior and is generally unknown. It can be expressed by the Jacobian matrix of the appropriate stress measure with respect to the corresponding (elasto-plastic) strain measure:

$$\mathbb{C}^\mathrm{ref} = \det(\boldsymbol{F}_\mathrm{p})\frac{\mathrm{d}\left(\boldsymbol{F}_\mathrm{p}^{-1}\,\boldsymbol{S}\,\boldsymbol{F}_\mathrm{p}^{-\mathrm{T}}\right)}{\mathrm{d}\boldsymbol{E}}, \tag{8}$$

$$\mathbb{C}^\mathrm{cur} = \frac{\mathrm{d}\boldsymbol{\sigma}}{\mathrm{d}\boldsymbol{\varepsilon}}, \tag{9}$$

$$\mathbb{C}^\mathrm{int} = \frac{\mathrm{d}\boldsymbol{P}}{\mathrm{d}\boldsymbol{F}}. \tag{10}$$

In Eq. (8) the 2nd Piola–Kirchhoff stress $\boldsymbol{S}$ must be transformed from the plastic configuration into the reference configuration using the plastic deformation gradient. The strain measures depend on the (total) deformation gradient $\boldsymbol{F}$ and are specified in analogy to the elastic strain measures, Eqs. (1)–(3):

$$\boldsymbol{E}(\boldsymbol{F}) := \frac{1}{2}(\boldsymbol{F}^\mathrm{T}\boldsymbol{F} - \boldsymbol{I}), \tag{11}$$

$$\boldsymbol{\varepsilon}_\mathrm{log}(\boldsymbol{F}) := \frac{1}{2}\ln(\boldsymbol{F}\boldsymbol{F}^\mathrm{T}), \tag{12}$$

$$\boldsymbol{\varepsilon}_\mathrm{EA}(\boldsymbol{F}) := \frac{1}{2}(\boldsymbol{I} - (\boldsymbol{F}\boldsymbol{F}^\mathrm{T})^{-1}). \tag{13}$$

The Green–Lagrange strain $\boldsymbol{E}$ is now defined within the reference configuration. Logarithmic ($\boldsymbol{\varepsilon}_\mathrm{log}$) and Euler–Almansi ($\boldsymbol{\varepsilon}_\mathrm{EA}$) strain are still defined within the current configuration.

There are several approaches available for the numerical differentiation of Eqs. (8)–(10). The complex-step derivative approximation approach is proved to be of second order accurate and does not suffer from roundoff errors in floating point arithmetics that limit the accuracy of the central- and forward-difference formula [17–20]. Since our methods, Section 5, are applicable to each of these approaches, we demonstrate their usage exemplarily by employing the forward-difference formula in accordance with [8–12]. The components of $\mathbb{C}^\mathrm{ref}$ and $\mathbb{C}^\mathrm{cur}$ can then be calculated as:

$$\mathbb{C}_{ijkl}^\mathrm{ref} \approx \frac{\det(\boldsymbol{F}_\mathrm{p})F_{\mathrm{p}\,im}^{-1}\left(S_{mn}(\boldsymbol{E}^{(kl)}) - S_{mn}(\boldsymbol{E})\right)F_{\mathrm{p}\,nj}^{-\mathrm{T}}}{\Delta E_{kl}^{(kl)}}, \tag{14}$$

with

$$\boldsymbol{E}^{(kl)} = \boldsymbol{E} + \Delta\boldsymbol{E}^{(kl)} = \boldsymbol{E} + \frac{\Delta E}{2}\left(\boldsymbol{G}_k \otimes \boldsymbol{G}_l + \boldsymbol{G}_l \otimes \boldsymbol{G}_k\right), \tag{15}$$

and

$$\mathbb{C}_{ijkl}^\mathrm{cur} = \frac{\mathrm{d}\sigma_{ij}}{\mathrm{d}\varepsilon_{kl}} \approx \frac{\sigma_{ij}(\boldsymbol{\varepsilon}^{(kl)}) - \sigma_{ij}(\boldsymbol{\varepsilon})}{\Delta\varepsilon_{kl}^{(kl)}}, \quad \text{with} \tag{16}$$

$$\boldsymbol{\varepsilon}^{(kl)} = \boldsymbol{\varepsilon} + \Delta\boldsymbol{\varepsilon}^{(kl)} = \boldsymbol{\varepsilon} + \frac{\Delta\varepsilon}{2}\left(\boldsymbol{g}_k \otimes \boldsymbol{g}_l + \boldsymbol{g}_l \otimes \boldsymbol{g}_k\right), \tag{17}$$

where $\boldsymbol{g}_k$ and $\boldsymbol{g}_l$ are the $k$th and $l$th Eulerian basis vectors, $\boldsymbol{G}_k$ and $\boldsymbol{G}_l$ are the $k$th and $l$th Lagrangian basis vectors, $k, l \in \{1, 2, 3\}$, $\Delta\boldsymbol{\varepsilon}^{(kl)}$ and $\Delta\boldsymbol{E}^{(kl)}$ are the strain perturbations in the tensor component $(kl)$, and $\Delta\varepsilon$ and $\Delta E$ are the

scalar perturbation magnitudes. Since the strain tensors $\boldsymbol{\varepsilon}$ and $\boldsymbol{E}$ are symmetric, only six perturbation steps are necessary in order to assemble $\mathbb{C}^{\text{ref}}$ resp. $\mathbb{C}^{\text{cur}}$. In contrast, nine perturbation steps are necessary to assemble $\mathbb{C}^{\text{int}}$ because in general the deformation gradient is not symmetric:

$$\mathbb{C}^{\text{int}}_{ijkl} \approx \frac{P_{ij}(\boldsymbol{F}^{(kl)}) - P_{ij}(\boldsymbol{F})}{\Delta F^{(kl)}_{kl}}, \quad \text{with} \tag{18}$$

$$\boldsymbol{F}^{(kl)} = \boldsymbol{F} + \Delta\boldsymbol{F}^{(kl)} = \boldsymbol{F} + \Delta F\, \boldsymbol{g}_k \otimes \boldsymbol{G}_l, \tag{19}$$

where $\Delta F$ is the perturbation magnitude in analogy to $\Delta E$ and $\Delta\varepsilon$.

Most sophisticated material models derive a stress response based on the deformation gradient $\boldsymbol{F}$. Therefore, in order to evaluate Eqs. (14) and (16), we have to calculate a perturbed deformation gradient $\boldsymbol{F}^{(kl)}$ such that a directed strain perturbation according to Eq. (15) or (17) is realized. As a consequence, for a strain perturbation within the reference configuration $\boldsymbol{F}^{(kl)}$ is defined by:

$$\Delta\boldsymbol{E}^{(kl)} \stackrel{!}{=} \boldsymbol{E}^{(kl)} - \boldsymbol{E} = \boldsymbol{E}(\boldsymbol{F}^{(kl)}) - \boldsymbol{E}(\boldsymbol{F}), \tag{20}$$

or for a strain perturbation within the current configuration:

$$\Delta\boldsymbol{\varepsilon}^{(kl)} \stackrel{!}{=} \boldsymbol{\varepsilon}^{(kl)} - \boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}(\boldsymbol{F}^{(kl)}) - \boldsymbol{\varepsilon}(\boldsymbol{F}). \tag{21}$$

Note, that $\boldsymbol{F}^{(kl)}$ in Eqs. (20) and (21) is not necessarily modified with respect to $\boldsymbol{F}$ only in its $kl$-component.

The perturbation within the intermediate configuration is specified in Eq. (19) and does not have to be evaluated. In this case, $\boldsymbol{F}^{(kl)}$ is modified with respect to $\boldsymbol{F}$ only in its $kl$-component:

$$\Delta\boldsymbol{F}^{(kl)} = \Delta F\, \boldsymbol{g}_k \otimes \boldsymbol{G}_l. \tag{22}$$

However, a solver requires the stress response and the elasto-plastic tangent stiffness tensor as output variables within a specific configuration. ABAQUS/STANDARD for example expects the elasto-plastic tangent stiffness tensor within the current configuration. As consequence, the tensor has to be transformed subsequently to the required configuration, if necessary.

## 4. Survey of existing methods

The survey was compiled to the best of the authors' knowledge and comprises widely spread and commonly used methods as well as a novel one. Nevertheless it does not claim completeness.

### *4.1. Tjahjanto (arbitrary strain)*

Tjahjanto [12] assembles the tangent stiffness tensor within the intermediate configuration, Eq. (18). Therefore one can directly perturb the deformation gradient following Eq. (22). As described before, nine perturbations are necessary to assemble $\mathbb{C}^{\text{int}}$. When working with ABAQUS/STANDARD as solver, the assembled Jacobian matrix must be transformed into the current configuration subsequently:

$$\mathbb{C}^{\text{cur}} = \text{Transformation}(\mathbb{C}^{\text{int}}, \boldsymbol{F}). \tag{23}$$

The demanding procedure of transformation is described in [12]. This method works irrespective of the used elastic strain measure as long as the stress response is transformed to the intermediate configuration, Eq. (6) or (7). In the evaluations presented in Section 6 we will exemplarily use the Green–Lagrange strain measure and label this method with *GL-Tjahjanto*.

### *4.2. Miehe (Green–Lagrange strain)*

Miehe [10] proposes a method applied to the Green–Lagrange strain measure, Eq. (11), in order to obtain a specific strain perturbation, Eq. (20). The perturbed deformation gradient $\boldsymbol{F}^{(kl)}$ is decomposed additively into an unperturbed deformation gradient $\boldsymbol{F}$ and a perturbation $\Delta\boldsymbol{F}^{(kl)}$:

$$\boldsymbol{F}^{(kl)} = \boldsymbol{F} + \Delta\boldsymbol{F}^{(kl)}. \tag{24}$$

The perturbation has the form:

$$\Delta \boldsymbol{F}^{(kl)} = \frac{\Delta E}{2}\left[\boldsymbol{F}^{-\mathrm{T}} \cdot \boldsymbol{G}_k \otimes \boldsymbol{G}_l + \boldsymbol{F}^{-\mathrm{T}} \cdot \boldsymbol{G}_l \otimes \boldsymbol{G}_k\right]. \tag{25}$$

With this approach $\boldsymbol{E}^{(kl)} - \boldsymbol{E}$ will not result in a perfectly directed strain perturbation $\Delta \boldsymbol{E}^{(kl)}$ defined in Eq. (15). The tangent stiffness tensor is assembled using the finite difference scheme, Eq. (14). In the evaluations presented in Section 6 this method will be labeled with *GL-Miehe*. In [10] Miehe also derived a solution to calculate the perturbation for a strain measure within the current configuration. However, this method is not regarded in this paper.

### 4.3. Kalidindi (logarithmic strain)

Kalidindi [11] developed a method explicitly for the interaction with ABAQUS/STANDARD and thus conducts the perturbation directly within the current configuration by using the logarithmic strain measure, Eq. (12). To create a specific strain perturbation an additive decomposition of the perturbed deformation gradient is proposed:

$$\boldsymbol{F}^{(kl)}(t + \Delta t) = \boldsymbol{F}(t + \Delta t) + \Delta \boldsymbol{\varepsilon}^{(kl)}_{\log}(t + \Delta t)\boldsymbol{F}(t), \tag{26}$$

whereby $\boldsymbol{F}(t)$ arises from the latest converged equilibrium state, $\boldsymbol{F}(t + \Delta t)$ describes the deformation gradient at the current time increment and $\Delta \boldsymbol{\varepsilon}^{(kl)}_{\log}$ indicates the intended strain perturbation in accordance with Eq. (17). In case of small time steps, $\Delta t \to 0$, Eq. (26) can be simplified to:

$$\boldsymbol{F}^{(kl)} = \boldsymbol{F} + \Delta \boldsymbol{\varepsilon}^{(kl)}_{\log}\boldsymbol{F}. \tag{27}$$

As Eq. (27) provides slightly better results than Eq. (26), we use this form for the evaluation in Section 6 and label it with *LOG-Kalidindi*. However, one must keep in mind that this approach does not fulfill Eq. (21) with respect to the definition of $\Delta \boldsymbol{\varepsilon}^{(kl)}$ in Eq. (17) exactly.

### 4.4. Taxer (arbitrary strain)

Taxer [13] derived a direct way to calculate the tangent stiffness tensor based on Eq. (8) or (9). For six nearly arbitrary strain perturbations, with the only restriction that the strain perturbations have to be independent among themselves and must not result in a rigid body rotation which would result in zero strain difference otherwise, the stress response is determined. In case of working within the current configuration, for example, the corresponding six instances of Eq. (9) are condensed and rearranged as shown in Eq. (28):

$$\begin{aligned} & \mathbb{C}^{\mathrm{cur}}_{[3\times3\times3\times3]} : \Delta \boldsymbol{\varepsilon}_{[3\times3]} = \Delta \boldsymbol{\sigma}_{[3\times3]} \\ \Rightarrow \quad & \Delta \boldsymbol{\varepsilon}_{[36\times21]} \cdot \mathbb{C}^{\mathrm{cur}}_{[21\times1]} = \Delta \boldsymbol{\sigma}_{[36\times1]}, \end{aligned} \tag{28}$$

where the subscript in square brackets indicates the dimension of the particular matrix. The symmetry of $\mathbb{C}^{\mathrm{cur}}$ allows its dimensional reduction to a matrix of $[21 \times 1]$. $\Delta \boldsymbol{\varepsilon}_{[36\times21]}$ and $\Delta \boldsymbol{\sigma}_{[36\times1]}$ contain the concatenation of the strain and stress differences caused by the six perturbation steps. For clarification an excerpt of Eq. (28) is detailed in Eq. (29), where an upper index $(i)$, $i = 1, 2, \ldots, 6$, refers to the different perturbations.

$$\begin{pmatrix} \Delta\varepsilon^{(1)}_{11} & \Delta\varepsilon^{(1)}_{22} & \Delta\varepsilon^{(1)}_{33} & \Delta\varepsilon^{(1)}_{12} & \Delta\varepsilon^{(1)}_{13} & \Delta\varepsilon^{(1)}_{23} & 0 & \ldots \\ 0 & \Delta\varepsilon^{(1)}_{11} & 0 & 0 & 0 & 0 & \Delta\varepsilon^{(1)}_{22} & \ldots \\ 0 & 0 & \Delta\varepsilon^{(1)}_{11} & 0 & 0 & 0 & 0 & \ldots \\ 0 & 0 & 0 & \Delta\varepsilon^{(1)}_{11} & 0 & 0 & 0 & \ldots \\ 0 & 0 & 0 & 0 & \Delta\varepsilon^{(1)}_{11} & 0 & 0 & \ldots \\ 0 & 0 & 0 & 0 & 0 & \Delta\varepsilon^{(1)}_{11} & 0 & \ldots \\ \varepsilon^{(2)}_{11} & \Delta\varepsilon^{(2)}_{22} & \Delta\varepsilon^{(2)}_{33} & \Delta\varepsilon^{(2)}_{12} & \Delta\varepsilon^{(2)}_{13} & \Delta\varepsilon^{(2)}_{23} & 0 & \ldots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \\ 0 & 0 & 0 & 0 & 0 & \Delta\varepsilon^{(6)}_{11} & 0 & \ldots \end{pmatrix} \cdot \begin{pmatrix} C^{\mathrm{cur}}_{11} \\ C^{\mathrm{cur}}_{12} \\ C^{\mathrm{cur}}_{13} \\ C^{\mathrm{cur}}_{14} \\ C^{\mathrm{cur}}_{15} \\ C^{\mathrm{cur}}_{16} \\ C^{\mathrm{cur}}_{22} \\ \vdots \\ C^{\mathrm{cur}}_{66} \end{pmatrix} = \begin{pmatrix} \Delta\sigma^{(1)}_{11} \\ \Delta\sigma^{(1)}_{22} \\ \Delta\sigma^{(1)}_{33} \\ \Delta\sigma^{(1)}_{12} \\ \Delta\sigma^{(1)}_{13} \\ \Delta\sigma^{(1)}_{23} \\ \Delta\sigma^{(2)}_{11} \\ \vdots \\ \Delta\sigma^{(6)}_{23} \end{pmatrix}. \tag{29}$$

As Eq. (28) represents an overdetermined system of linear equations, it can be transformed into an equalization problem. Finally we obtain $\mathbb{C}^{\text{cur}}$ from:

$$\mathbb{C}^{\text{cur}}_{[21\times 1]} = \left(\Delta\boldsymbol{\varepsilon}^{\text{T}}_{[36\times 21]}\Delta\boldsymbol{\varepsilon}_{[36\times 21]}\right)^{-1}\Delta\boldsymbol{\varepsilon}^{\text{T}}_{[36\times 21]}\Delta\boldsymbol{\sigma}_{[36\times 1]}. \tag{30}$$

The tangent stiffness tensor, which is required by ABAQUS/STANDARD with the dimension [6 × 6], can be easily constructed by rearranging the components of $\mathbb{C}^{\text{cur}}_{[21\times 1]}$. This method can be used in combination with any strain measure within the current configuration as well as within the reference configuration:

$$\mathbb{C}^{\text{ref}}_{[21\times 1]} = \left(\Delta\boldsymbol{E}^{\text{T}}_{[36\times 21]}\Delta\boldsymbol{E}_{[36\times 21]}\right)^{-1}\Delta\boldsymbol{E}^{\text{T}}_{[36\times 21]}\Delta\boldsymbol{S}_{[36\times 1]}. \tag{31}$$

In the evaluations presented in Section 6 we will exemplarily use the logarithmic strain measure and label this method with *LOG-Taxer*.

*Modification*

In order to save computational effort the authors propose a slightly modified approach. While rearranging the approximation of $\mathbb{C}^{\text{cur}}$ we do not utilize its full symmetry:

$$\begin{aligned} &\mathbb{C}^{\text{cur}}_{[3\times 3\times 3\times 3]} : \Delta\boldsymbol{\varepsilon}_{[3\times 3]} = \Delta\boldsymbol{\sigma}_{[3\times 3]} \\ \Rightarrow\quad &\Delta\boldsymbol{\varepsilon}_{[36\times 36]}\cdot\mathbb{C}^{\text{cur}}_{[36\times 1]} = \Delta\boldsymbol{\sigma}_{[36\times 1]}. \end{aligned} \tag{32}$$

The tangent stiffness tensor within the current configuration can then be computed directly:

$$\mathbb{C}^{\text{cur}}_{[36\times 1]} = \Delta\boldsymbol{\varepsilon}^{-1}_{[36\times 36]}\Delta\boldsymbol{\sigma}_{[36\times 1]}. \tag{33}$$

Within the reference configuration the equation has the form:

$$\mathbb{C}^{\text{ref}}_{[36\times 1]} = \Delta\boldsymbol{E}^{-1}_{[36\times 36]}\Delta\boldsymbol{S}_{[36\times 1]}. \tag{34}$$

The accuracy of approximating $\mathbb{C}^{\text{cur}}_{[36\times 1]}$ or $\mathbb{C}^{\text{ref}}_{[36\times 1]}$ is the very same as it is for the original approach, Eq. (30) or (31). Again, $\mathbb{C}_{[6\times 6]}$ can be easily constructed by rearranging the components of $\mathbb{C}_{[36\times 1]}$.

## 5. Methods proposed by the authors

Our approach is aimed to ensure that the strain perturbation occurs exactly for only one strain component as defined in Eqs. (15) and (17), hence the tangent stiffness tensor can be formed easily as presented in Eqs. (14) and (16). The error generated during the assembly is the smaller the preciser the strain perturbation is conducted. For each strain measure an individual derivation is necessary in order to generate an exact strain perturbation. The proposed methods solve Eqs. (20) and (21) for the perturbed deformation gradient $\boldsymbol{F}^{(kl)}$ either pursuing a direct or an iterative approach.

### 5.1. Green–Lagrange strain

#### 5.1.1. Direct approach

First we substitute the strain in Eq. (20) with the Green–Lagrange strain measure defined in Eq. (11). For a given unperturbed deformation gradient $\boldsymbol{F}$ and a specific choice of $\Delta\boldsymbol{E}^{(kl)}$, Eq. (20) can then be solved directly for the sought perturbed deformation gradient $\boldsymbol{F}^{(kl)}$ as follows:

$$\begin{aligned} \Delta\boldsymbol{E}^{(kl)} &= \frac{1}{2}\left[\boldsymbol{F}^{(kl)^{\text{T}}}\boldsymbol{F}^{(kl)} - \boldsymbol{I}\right] - \frac{1}{2}\left[\boldsymbol{F}^{\text{T}}\boldsymbol{F} - \boldsymbol{I}\right] \\ \boldsymbol{F}^{(kl)^{\text{T}}}\boldsymbol{F}^{(kl)} &= 2\Delta\boldsymbol{E}^{(kl)} + \boldsymbol{F}^{\text{T}}\boldsymbol{F} =: \boldsymbol{A}. \end{aligned} \tag{35}$$

We obtain one specific analytic solution for the perturbed deformation gradient $\boldsymbol{F}^{(kl)}$ by conducting a spectral decomposition of $\boldsymbol{A}$ based on the eigenvectors $\boldsymbol{n}_{\mathrm{A}}^{j}$ and eigenvalues $\lambda_{\mathrm{A}}^{j},\ j = 1, \ldots, 3$:

$$\boxed{\boldsymbol{F}^{(kl)} = \sum_{j=1}^{3} \sqrt{\lambda_A^j}\, \boldsymbol{n}_A^j \otimes \boldsymbol{n}_A^j}. \tag{36}$$

In the evaluations presented in Section 6 this method will be labeled with *GL-Direct*.

### 5.1.2. Iterative approach

Similar to Miehe, Section 4.2, and Kalidindi, Section 4.3, the perturbed deformation gradient $\boldsymbol{F}^{(kl)}$ is now additively composed of the perturbation $\Delta\boldsymbol{F}^{(kl)}$ and the unperturbed deformation gradient $\boldsymbol{F}$:

$$\boldsymbol{F}^{(kl)} = \boldsymbol{F} + \Delta\boldsymbol{F}^{(kl)}, \tag{37}$$

Eq. (20) can be solved for the sought perturbation $\Delta\boldsymbol{F}^{(kl)}$ as follows:

$$\Delta\boldsymbol{E}^{(kl)} = \frac{1}{2}\left[(\boldsymbol{F} + \Delta\boldsymbol{F}^{(kl)})^{\mathrm{T}}(\boldsymbol{F} + \Delta\boldsymbol{F}^{(kl)}) - \boldsymbol{I}\right] - \frac{1}{2}\left[\boldsymbol{F}^{\mathrm{T}}\boldsymbol{F} - \boldsymbol{I}\right]$$
$$\Delta\boldsymbol{E}^{(kl)} - \frac{1}{2}\Delta\boldsymbol{F}^{(kl)^{\mathrm{T}}}\Delta\boldsymbol{F}^{(kl)} = \frac{1}{2}\left[\Delta\boldsymbol{F}^{(kl)^{\mathrm{T}}}\boldsymbol{F} + \left(\Delta\boldsymbol{F}^{(kl)^{\mathrm{T}}}\boldsymbol{F}\right)^{\mathrm{T}}\right].$$

Since the perturbation magnitude is commonly small, $\Delta\boldsymbol{F}^{(kl)^{\mathrm{T}}}\Delta\boldsymbol{F}^{(kl)}$ can be neglected in a first approximation.

$$\Delta\boldsymbol{E}^{(kl)} - \frac{1}{2}\underbrace{\Delta\boldsymbol{F}^{(kl)^{\mathrm{T}}}\Delta\boldsymbol{F}^{(kl)}}_{\approx 0} \approx \mathrm{sym}(\Delta\boldsymbol{F}^{(kl)^{\mathrm{T}}}\boldsymbol{F}). \tag{38}$$

Assuming that $\Delta\boldsymbol{F}^{(kl)^{\mathrm{T}}}\boldsymbol{F}$ itself is symmetric, one possible solution is $\Delta\boldsymbol{F}^{(kl)^{\mathrm{T}}}\boldsymbol{F} \approx \Delta\boldsymbol{E}^{(kl)}$. Solving for the perturbation, one gets:

$$\Delta\boldsymbol{F}^{(kl)} \approx \Delta\boldsymbol{E}^{(kl)}\boldsymbol{F}^{-1}.$$

The value of the perturbation $\Delta\boldsymbol{F}^{(kl)}$ can now be optionally used to update the term $\Delta\boldsymbol{F}^{(kl)^{\mathrm{T}}}\Delta\boldsymbol{F}^{(kl)}$ in Eq. (38) in order to increase the accuracy of the method by one or more iterations. Without iterating, we obtain from Eq. (37) one specific solution for the perturbed deformation gradient:

$$\boxed{\boldsymbol{F}^{(kl)} \approx \boldsymbol{F} + \Delta\boldsymbol{E}^{(kl)}\boldsymbol{F}^{-1}}. \tag{39}$$

In the evaluations presented in Section 6 the approach with no additional iteration will be labeled with *GL-Iter 0* and the approach with one additional iteration with *GL-Iter 1*.

## 5.2. Logarithmic strain

### 5.2.1. Direct approach

This procedure has already been presented in [21] in a modified form, where a multiplicative decomposition of the perturbed deformation gradient was chosen. Here, we present a direct approach with reduced computational effort. We substitute the strain in Eq. (21) with the logarithmic strain measure defined in Eq. (12), which is the strain measure used by ABAQUS/STANDARD. For a given $\boldsymbol{F}$ and a specific choice of $\Delta\boldsymbol{\varepsilon}_{\log}^{(kl)}$, Eq. (21) can then be solved for the sought perturbed deformation gradient $\boldsymbol{F}^{(kl)}$ as follows:

$$\Delta\boldsymbol{\varepsilon}_{\log}^{(kl)} = \frac{1}{2}\ln\left(\boldsymbol{F}^{(kl)}\boldsymbol{F}^{(kl)^{\mathrm{T}}}\right) - \frac{1}{2}\ln\left(\boldsymbol{F}\boldsymbol{F}^{\mathrm{T}}\right)$$
$$\ln\left(\boldsymbol{F}^{(kl)}\boldsymbol{F}^{(kl)^{\mathrm{T}}}\right) = 2\Delta\boldsymbol{\varepsilon}_{\log}^{(kl)} + \ln\left(\boldsymbol{F}\boldsymbol{F}^{\mathrm{T}}\right) =: \boldsymbol{A}.$$

We obtain one specific analytic solution for the perturbed deformation gradient $\boldsymbol{F}^{(kl)}$ by conducting a spectral decomposition of $\boldsymbol{A}$ based on the eigenvectors $\boldsymbol{n}_{\mathrm{A}}^{j}$ and eigenvalues $\lambda_{\mathrm{A}}^{j},\ j = 1, \ldots, 3$:

$$\boxed{\boldsymbol{F}^{(kl)} = \sum_{j=1}^{3} \sqrt{\mathrm{e}^{\lambda_{\mathrm{A}}^{j}}}\, \boldsymbol{n}_{\mathrm{A}}^{j} \otimes \boldsymbol{n}_{\mathrm{A}}^{j}}. \tag{40}$$

In the evaluations presented in Section 6 this method will be labeled with *LOG-Direct*.

### 5.2.2. Iterative approach

Here we use the additive decomposition, Eq. (37), and solve Eq. (21) for the logarithmic strain measure:

$$\Delta\boldsymbol{\varepsilon}_{\log}^{(kl)} = \frac{1}{2}\ln\left(\left(\boldsymbol{F} + \Delta\boldsymbol{F}^{(kl)}\right)\left(\boldsymbol{F} + \Delta\boldsymbol{F}^{(kl)}\right)^{\mathrm{T}}\right) - \frac{1}{2}\ln\left(\boldsymbol{F}\boldsymbol{F}^{\mathrm{T}}\right)$$

$$\left(\boldsymbol{F} + \Delta\boldsymbol{F}^{(kl)}\right)\left(\boldsymbol{F} + \Delta\boldsymbol{F}^{(kl)}\right)^{\mathrm{T}} = \underbrace{\exp\Bigg(\underbrace{2\Delta\boldsymbol{\varepsilon}_{\log}^{(kl)} + \ln\left(\boldsymbol{F}\boldsymbol{F}^{\mathrm{T}}\right)}_{\boldsymbol{A}}\Bigg)}_{\boldsymbol{B}}.$$

The equation can be expanded to the form:

$$\boldsymbol{F}\boldsymbol{F}^{\mathrm{T}} + \Delta\boldsymbol{F}^{(kl)}\boldsymbol{F}^{\mathrm{T}} + \boldsymbol{F}\Delta\boldsymbol{F}^{(kl)^{\mathrm{T}}} + \Delta\boldsymbol{F}^{(kl)}\Delta\boldsymbol{F}^{(kl)^{\mathrm{T}}} = \boldsymbol{B}$$

and then be rewritten as:

$$\frac{1}{2}\left[\boldsymbol{B} - \boldsymbol{F}\boldsymbol{F}^{\mathrm{T}} - \Delta\boldsymbol{F}^{(kl)}\Delta\boldsymbol{F}^{(kl)^{\mathrm{T}}}\right] = \frac{1}{2}\left[\Delta\boldsymbol{F}^{(kl)}\boldsymbol{F}^{\mathrm{T}} + \boldsymbol{F}\Delta\boldsymbol{F}^{(kl)^{\mathrm{T}}}\right].$$

Again neglecting the term $\Delta\boldsymbol{F}^{(kl)}\Delta\boldsymbol{F}^{(kl)^{\mathrm{T}}}$ one obtains:

$$\frac{1}{2}\left[\boldsymbol{B} - \boldsymbol{F}\boldsymbol{F}^{\mathrm{T}} - \underbrace{\Delta\boldsymbol{F}^{(kl)}\Delta\boldsymbol{F}^{(kl)^{\mathrm{T}}}}_{\approx 0}\right] \approx \mathrm{sym}(\Delta\boldsymbol{F}^{(kl)}\boldsymbol{F}^{\mathrm{T}}). \tag{41}$$

Assuming that $\Delta\boldsymbol{F}^{(kl)}\boldsymbol{F}^{\mathrm{T}}$ itself is symmetric, one possible solution is $\Delta\boldsymbol{F}^{(kl)}\boldsymbol{F}^{\mathrm{T}} \approx \frac{1}{2}\left[\boldsymbol{B} - \boldsymbol{F}\boldsymbol{F}^{\mathrm{T}}\right]$. Solved for the perturbation, one gets:

$$\Delta\boldsymbol{F}^{(kl)} \approx \frac{1}{2}\left[\boldsymbol{B} - \boldsymbol{F}\boldsymbol{F}^{\mathrm{T}}\right]\boldsymbol{F}^{-\mathrm{T}}.$$

The value of the perturbation $\Delta\boldsymbol{F}^{(kl)}$ can now be used to update the term $\Delta\boldsymbol{F}^{(kl)}\Delta\boldsymbol{F}^{(kl)^{\mathrm{T}}}$ in Eq. (41) iteratively in order to increase the accuracy. Without additional iterations we obtain one specific solution for the perturbed deformation gradient:

$$\boxed{\boldsymbol{F}^{(kl)} \approx \boldsymbol{F} + \frac{1}{2}\left[\boldsymbol{B} - \boldsymbol{F}\boldsymbol{F}^{\mathrm{T}}\right]\boldsymbol{F}^{-\mathrm{T}}} \tag{42}$$

whereby $\boldsymbol{B}$ can be determined by conducting a spectral decomposition of $\boldsymbol{A}$ based on the eigenvectors $\boldsymbol{n}_{\mathrm{A}}^{j}$ and eigenvalues $\lambda_{\mathrm{A}}^{j},\ j = 1, \ldots, 3$:

$$\boldsymbol{B} = \sum_{j=1}^{3} \mathrm{e}^{\lambda_{\mathrm{A}}^{j}}\boldsymbol{n}_{\mathrm{A}}^{j} \otimes \boldsymbol{n}_{\mathrm{A}}^{j}.$$

In the evaluations presented in Section 6 the approach with no additional iteration will be labeled with *LOG-Iter 0* and the approach with one additional iteration with *LOG-Iter 1*.

### 5.3. Euler–Almansi strain

#### 5.3.1. Direct approach

First we substitute the strain in Eq. (21) with the Euler–Almansi strain measure defined in Eq. (13). For a given unperturbed deformation gradient $\boldsymbol{F}$ and a specific choice of $\Delta\boldsymbol{\varepsilon}_{\mathrm{EA}}^{(kl)}$, Eq. (21) can then be solved directly for the sought perturbed deformation gradient $\boldsymbol{F}^{(kl)}$ as follows:

$$\Delta\boldsymbol{\varepsilon}_{\mathrm{EA}}^{(kl)} = \frac{1}{2}\left[\boldsymbol{I} - \left(\boldsymbol{F}^{(kl)}\boldsymbol{F}^{(kl)^{\mathrm{T}}}\right)^{-1}\right] - \frac{1}{2}\left[\boldsymbol{I} - \left(\boldsymbol{F}\boldsymbol{F}^{\mathrm{T}}\right)^{-1}\right]$$

$$\left(\boldsymbol{F}^{(kl)}\boldsymbol{F}^{(kl)^{\mathrm{T}}}\right)^{-1} = \left(\boldsymbol{F}\boldsymbol{F}^{\mathrm{T}}\right)^{-1} - 2\Delta\boldsymbol{\varepsilon}_{\mathrm{EA}}^{(kl)}$$

$$\boldsymbol{F}^{(kl)}\boldsymbol{F}^{(kl)^{\mathrm{T}}} = \left[\left(\boldsymbol{F}\boldsymbol{F}^{\mathrm{T}}\right)^{-1} - 2\Delta\boldsymbol{\varepsilon}_{\mathrm{EA}}^{(kl)}\right]^{-1} =: \boldsymbol{A}.$$

We obtain one specific analytic solution for the perturbed deformation gradient $\boldsymbol{F}^{(kl)}$ by conducting a spectral decomposition of $\boldsymbol{A}$ based on the eigenvectors $\boldsymbol{n}_{\mathrm{A}}^{j}$ and eigenvalues $\lambda_{\mathrm{A}}^{j}$, $j = 1, \ldots, 3$:

$$\boxed{\Delta\boldsymbol{F}^{(kl)} = \sum_{j=1}^{3}\sqrt{\lambda_A^j}\,\boldsymbol{n}_A^j \otimes \boldsymbol{n}_A^j}. \tag{43}$$

In the evaluations presented in Section 6 this method will be labeled with *EA-Direct*.

#### 5.3.2. Iterative approach

We perform an additive decomposition of the perturbed deformation gradient $\boldsymbol{F}^{(kl)}$ and assume $\Delta\boldsymbol{F}^{(kl)}$ to be small. Then we can introduce a linearized approximation for the inverse of $\boldsymbol{F}^{(kl)}$ by means of auxiliary variables $\tilde{\boldsymbol{F}}$ and $\Delta\tilde{\boldsymbol{F}}^{(kl)}$, whereby $\tilde{\boldsymbol{F}}^{-1} \approx \boldsymbol{F}$ is assumed:

$$\boldsymbol{F}^{(kl)} = \boldsymbol{F} + \Delta\boldsymbol{F}^{(kl)} =: \left(\tilde{\boldsymbol{F}} + \Delta\tilde{\boldsymbol{F}}^{(kl)}\right)^{-1}. \tag{44}$$

Eq. (21) can then be solved for the sought perturbation $\Delta\boldsymbol{F}^{(kl)}$ as follows:

$$\Delta\boldsymbol{\varepsilon}_{\mathrm{EA}}^{(kl)} = \frac{1}{2}\left[\boldsymbol{I} - \left((\boldsymbol{F} + \Delta\boldsymbol{F}^{(kl)})(\boldsymbol{F} + \Delta\boldsymbol{F}^{(kl)})^{\mathrm{T}}\right)^{-1}\right] - \frac{1}{2}\left[\boldsymbol{I} - \left(\boldsymbol{F}\boldsymbol{F}^{\mathrm{T}}\right)^{-1}\right] \tag{45}$$

$$\Delta\boldsymbol{\varepsilon}_{\mathrm{EA}}^{(kl)} + \frac{1}{2}\Delta\tilde{\boldsymbol{F}}^{(kl)^{\mathrm{T}}}\Delta\tilde{\boldsymbol{F}}^{(kl)} = -\frac{1}{2}\left[\Delta\tilde{\boldsymbol{F}}^{(kl)^{\mathrm{T}}}\tilde{\boldsymbol{F}} + \left(\Delta\tilde{\boldsymbol{F}}^{(kl)^{\mathrm{T}}}\tilde{\boldsymbol{F}}\right)^{\mathrm{T}}\right].$$

In a first approximation we assume the term $\Delta\tilde{\boldsymbol{F}}^{(kl)^{\mathrm{T}}}\Delta\tilde{\boldsymbol{F}}^{(kl)}$ to vanish:

$$\Delta\boldsymbol{\varepsilon}_{\mathrm{EA}}^{(kl)} + \frac{1}{2}\underbrace{\Delta\tilde{\boldsymbol{F}}^{(kl)^{\mathrm{T}}}\Delta\tilde{\boldsymbol{F}}^{(kl)}}_{\approx 0} \approx -\mathrm{sym}(\Delta\tilde{\boldsymbol{F}}^{(kl)^{\mathrm{T}}}\tilde{\boldsymbol{F}}). \tag{46}$$

Assuming that $\Delta\tilde{\boldsymbol{F}}^{(kl)^{\mathrm{T}}}\tilde{\boldsymbol{F}}$ itself is symmetric, one possible solution is $\Delta\tilde{\boldsymbol{F}}^{(kl)^{\mathrm{T}}}\tilde{\boldsymbol{F}} \approx -\Delta\boldsymbol{\varepsilon}_{\mathrm{EA}}^{(kl)}$. Since the perturbation is small ($\tilde{\boldsymbol{F}}^{-1} \approx \boldsymbol{F}$), we can write:

$$\Delta\tilde{\boldsymbol{F}}^{(kl)^{\mathrm{T}}} \approx -\Delta\boldsymbol{\varepsilon}_{\mathrm{EA}}^{(kl)}\boldsymbol{F}.$$

The value of the perturbation $\Delta\tilde{\boldsymbol{F}}^{(kl)^{\mathrm{T}}}$ can now be used to update the term $\Delta\tilde{\boldsymbol{F}}^{(kl)^{\mathrm{T}}}\Delta\tilde{\boldsymbol{F}}^{(kl)}$ in Eq. (46) iteratively in order to increase the accuracy. Inserting $\Delta\tilde{\boldsymbol{F}}^{(kl)}$ into Eq. (44) and substituting $\tilde{\boldsymbol{F}}$ by $\boldsymbol{F}^{-1}$ we obtain one specific solution for the perturbed deformation gradient without performing additional iterations:

$$\boxed{\boldsymbol{F}^{(kl)} \approx \left(\boldsymbol{F}^{-1} - \boldsymbol{F}^{\mathrm{T}}\Delta\boldsymbol{\varepsilon}_{\mathrm{EA}}^{(kl)^{\mathrm{T}}}\right)^{-1}}. \tag{47}$$

In the evaluations presented in Section 6 the approach with no additional iteration will be labeled with *EA-Iter 0* and the approach with one additional iteration with *EA-Iter 1*.

## 6. Evaluation of the methods

In order to assess as objectively as possible our methods, we apply them and the methods presented in Section 4 to a set of test simulations. A first analysis, Section 6.1, is dedicated to the sensitivity of the methods to the order of the perturbation magnitude. In Section 6.2 all methods' ability to determine the correct tangent stiffness tensor is compared with each other on the basis of a simple but expressive load case. The effect on the rate of convergence is studied in Section 6.3.

### 6.1. Sensitivity to the perturbation magnitude

The methods proposed by Miehe, Section 4.2, Kalidindi, Section 4.3 and the authors, Section 5, aim at an as precisely as possible directed strain perturbation. In order to determine the methods' accuracy with respect to the perturbation magnitude $\Delta e$ resp. $\Delta E$, for each method a perturbed deformation gradient $\boldsymbol{F}^{(kl)}$ is calculated based on the same artificial deformation gradient $\boldsymbol{F}$ with random values of its diagonal elements in the range of [0.5, 1.5] and [−0.5, 0.5] for its off-diagonal elements:

$$\boldsymbol{F} = \begin{pmatrix} [0.5, 1.5] & [-0.5, 0.5] & [-0.5, 0.5] \\ [-0.5, 0.5] & [0.5, 1.5] & [-0.5, 0.5] \\ [-0.5, 0.5] & [-0.5, 0.5] & [0.5, 1.5] \end{pmatrix}. \tag{48}$$

The perturbation magnitude is varied within a range of $10^{-15}$–$10^{-1}$. Depending on the strain measure the relative discrepancy $\boldsymbol{\Pi}^{(kl)}$ between the calculated and the intended strain perturbation can be determined as follows, cf. Eqs. (20) and (21):

$$\boldsymbol{\Pi}^{(kl)} = \left| \frac{\boldsymbol{\varepsilon}(\boldsymbol{F}^{(kl)}) - \boldsymbol{\varepsilon}(\boldsymbol{F})}{\Delta\varepsilon} - \frac{1}{2}\left(\boldsymbol{g}_k \otimes \boldsymbol{g}_l + \boldsymbol{g}_l \otimes \boldsymbol{g}_k\right) \right|, \tag{49}$$

$$\boldsymbol{\Pi}^{(kl)} = \left| \frac{\boldsymbol{E}(\boldsymbol{F}^{(kl)}) - \boldsymbol{E}(\boldsymbol{F})}{\Delta E} - \frac{1}{2}\left(\boldsymbol{G}_k \otimes \boldsymbol{G}_l + \boldsymbol{G}_l \otimes \boldsymbol{G}_k\right) \right|. \tag{50}$$

$\boldsymbol{\Pi}^{(kl)}$ shall be near to zero in all its components. Furthermore we distinguish between the relative error $\Pi^{\text{magnitude}}$, derived from the components pointing in the direction of perturbation and averaged over all six perturbations,

$$\Pi^{\text{magnitude}} = \frac{1}{6}\sum_{k=1}^{3}\sum_{l=k}^{3}\frac{1}{2}\left(\Pi_{kl}^{(kl)} + \Pi_{lk}^{(kl)}\right),$$

and the relative error $\Pi^{\text{direction}}$ derived from all the other components not pointing in the direction of perturbation and averaged over all six perturbations, Fig. 2. These two parameters can be interpreted as measures for the ability to generate the correct magnitude of the perturbation ($\Pi^{\text{magnitude}}$) and the ability to conduct the perturbation into the correct direction ($\Pi^{\text{direction}}$). Firstly, we observe that $\Pi^{\text{direction}}$ and $\Pi^{\text{magnitude}}$ evolve nearly identical irrespective of the applied method. Therefore the following observations are valid equally for the upper and the lower figure. For high perturbation magnitudes the direct approaches (... - *Direct*) possess the smallest error of all methods. But we have to be aware that depending on the initial deformation gradient there exists a maximum allowable perturbation magnitude to which a perturbation can be performed. This restriction is due to the fact that there are limits for the Green–Lagrange and the Euler–Almansi strain measure. The minimum Green–Lagrange strain ($\boldsymbol{F} \to 0$) is equal to −50% and the maximum Euler–Almansi strain ($\boldsymbol{F} \to \infty$) is equal to 50%. For *GL-Direct* we see that the error increases abruptly as soon as the strain perturbation reaches 10%. At this point a few of the 100 randomly generated deformation gradients in combination with the high strain perturbation lead to a tensor $\boldsymbol{A}$ in Eq. (35) which is not positive definite anymore (at least one negative eigenvalue). Beside this anomaly the error of the direct derivations increases while decreasing the perturbation magnitude due to the limits of computational accuracy. This is also the case for all the other methods using an additive decomposition as soon as the perturbation magnitude is smaller than

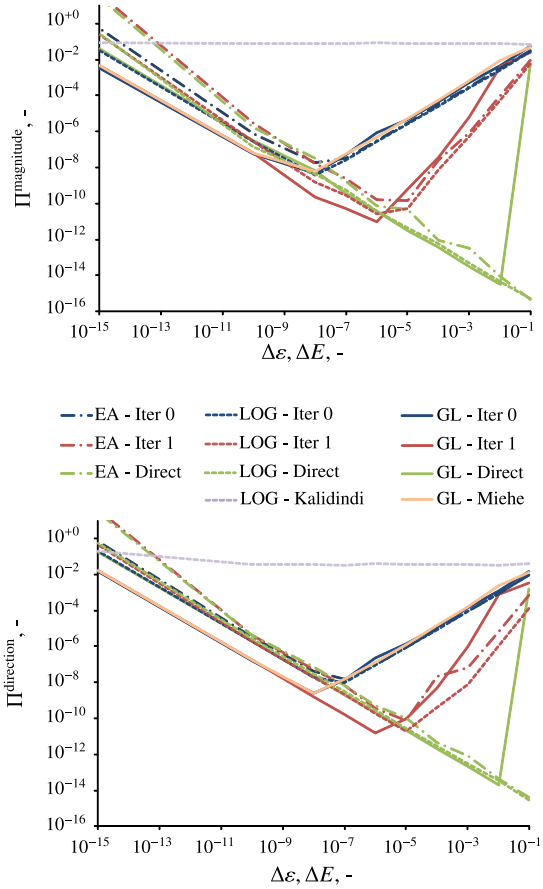


Fig. 2. Relative error $\Pi^{\text{magnitude}}$ and $\Pi^{\text{direction}}$ evaluated for perturbation magnitudes $10^{-15}$ and $10^{-10}, 10^{-9}, \ldots, 10^{-1}$. The graphs arise from 100 single runs with random deformation gradients according to Eq. (48).

a certain method dependent value. The method presented by Miehe, *GL-Miehe* and the method labeled with *GL-Iter 0* behave nearly identical. Together with the other approaches with additive decomposition and no additional iteration *EA-Iter 0* and *LOG-Iter 0* they have their minimum near a perturbation magnitude of $10^{-8}$. Conducting an additional iteration (... *-Iter 1*) decreases the error by several orders, especially for larger perturbation magnitudes. The minimum moves to a perturbation magnitude between $10^{-6}$ and $10^{-5}$. Kalidindi's method shows no dependency on the perturbation magnitude but possesses the highest error in magnitude and direction.

### 6.2. Sensitivity to the amount of deformation and rotation

In order to compare the presented methods with respect to the accuracy of the determined tangent stiffness tensor, we set up a simple test case consisting of a unit cell (dimensions: $1 \times 1 \times 1$) which is compressed by 10% in $x$-direction, sheared symmetrically by an angle of 52° and rotated by 180° about the $z$-axis subsequently, cf. Fig. 3. The unit cell consists of one fully integrated linear element with eight integration points, whereby the assessment bases on only one single integration point. The model describes a purely isotropic elastic material behavior (Young's modulus $E = 69$ GPa and Poisson's ratio $\nu = 0.33$), thus $\boldsymbol{F}_{\text{p}} = \boldsymbol{I}$ and $\boldsymbol{F} = \boldsymbol{F}_{\text{e}}$. The problem was solved utilizing the ABAQUS/STANDARD solver in combination with a user-defined material subroutine which calculates simultaneously the analytically correct tangent stiffness tensor $\tilde{\mathbb{C}}(E, \nu)$ and the tangent stiffness tensors $\mathbb{C}$ according to the presented methods. The perturbation magnitudes $\Delta E$, $\Delta\varepsilon$ and $\Delta F$ are set to $10^{-6}$. Only the analytically correct $\tilde{\mathbb{C}}$ is passed to ABAQUS/STANDARD, thus each method is evaluated along the exact same deformation path.

The difference between the numerically calculated $\mathbb{C}$ and the analytically correct tangent stiffness $\tilde{\mathbb{C}}$ is evaluated by two measures, Eq. (51): an error $\Pi^{\text{filled}}$ averaged among the components colored in red (dark) where the components of

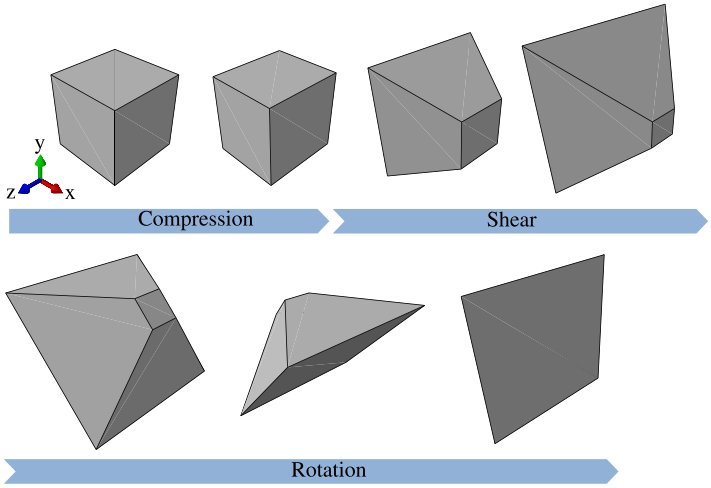


Fig. 3. Simulated deformation/rotation of a unit cell which is first compressed by 10% in $x$-direction, followed by a shear deformation and a rotation by 180° about the $z$-axis.

$\tilde{\mathbb{C}}$ are nonzero and an error $\Pi^{\text{unfilled}}$ averaged among the components colored in green (bright) where the components of $\tilde{\mathbb{C}}$ are zero:

$$\frac{\left|\tilde{\mathbb{C}} - \mathbb{C}\right|}{E} = \begin{pmatrix} \square & \square & \square & \square & \square & \square \\ \square & \square & \square & \square & \square & \square \\ \square & \square & \square & \square & \square & \square \\ \square & \square & \square & \square & \square & \square \\ \square & \square & \square & \square & \square & \square \\ \square & \square & \square & \square & \square & \square \end{pmatrix}. \tag{51}$$

Both types of error are normalized by Young's modulus $E$. Fig. 4 indicates each method's behavior over the loading path. For the method proposed by Kalidindi as well as for the method proposed by Tjahjanto the error increases as the deformation progresses. Especially when the shear deformation starts, the error $\Pi^{\text{unfilled}}$ increases by four to five orders of magnitude. This is due to the fact that the deformation gradient starts to become an unsymmetrical dense matrix. During rotation a slight dependency on the rotation angle is identifiable. In contrast, Miehe's approach shows no dependency on the amount of deformation and rotation. The error remains constant at around $10^{-7}$. Irrespective of the amount of deformation and rotation the errors $\Pi^{\text{filled}}$ and $\Pi^{\text{unfilled}}$ are in the range of $10^{-11}$–$10^{-9}$ for the methods proposed by the authors and in the range of $10^{-12}$–$10^{-11}$ for the approach proposed by Taxer. The direct approaches (... - *Direct*) and the method proposed by Taxer are only limited by the computational accuracy.

### 6.3. *Effect on the rate of convergence*

The unit cell from Section 6.2 is now discretized by $5^3$ fully integrated linear elements, Fig. 5. The elastic material behavior from before is extended by classical $J_2$-plasticity for isotropic hardening. As power law relationship between the stress and the amount of plastic strain we employed Ludwik's equation:

$$\sigma = \sigma_{\text{y}} + K \left|\varepsilon^{\text{p}}\right|^m, \tag{52}$$

with $\sigma_{\text{y}} = 200$ MPa, $K = 300$ MPa and $m = 0.25$. The bottom surface, $y = 0$, is fixed against displacement and on the top surface, $y = 1$, a Dirichlet boundary condition is applied ($u_x = -0.4$ and $u_y = -0.1$). To study the effect of the method used for the calculation of the tangent stiffness tensor on the rate of convergence, now the numerically calculated $\mathbb{C}$ is passed to ABAQUS/STANDARD. Since a comparative assessment of the accuracy of the authors' methods has already been presented in Sections 6.1 and 6.2, cf. Figs. 2 and 4, we restrict this study to the direct approaches. The total number of the required iterations is depicted in Table 1. Employing the Green–Lagrange strain measure, we observe the same number of required iterations for *GL-Miehe* and *GL-Direct*, although Fig. 4

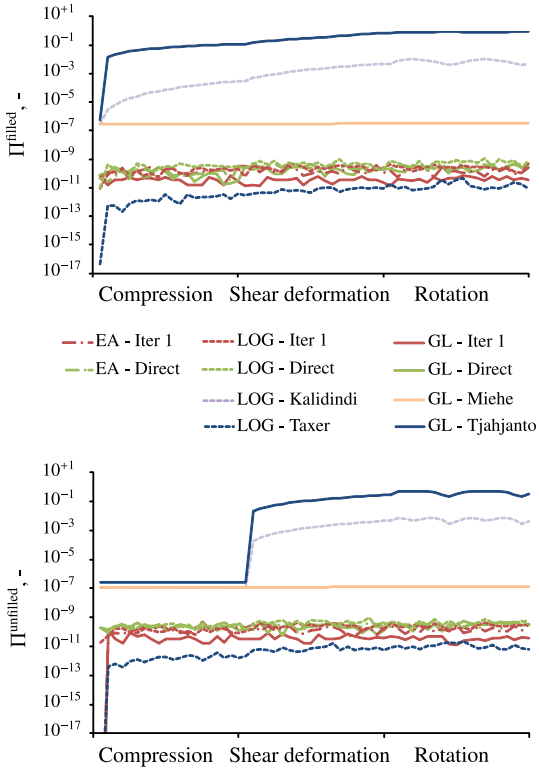


Fig. 4. Error evolution for all methods along the load path shown in Fig. 3. Each graph comprises 60 evenly distributed calculated points.

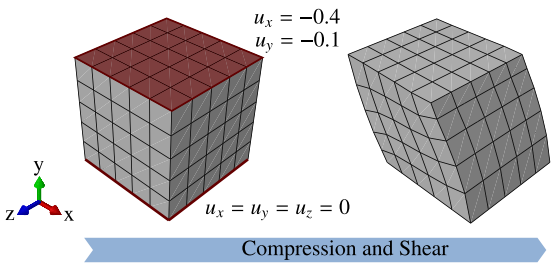


Fig. 5. Compression and shear deformation applied to a unit cell.

Table 1
Total number of required iterations.

| | | | |
|---|---|---|---|
| GL-Direct: | 137 | LOG-Direct: | 273 |
| GL-Miehe: | 137 | LOG-Kalidindi: | 292 |
| GL-Tjahjanto: | >3000 | LOG-Taxer: | 273 |

shows that the accuracy of *GL-Direct* is higher by three orders of magnitude. In this case the accuracy of the tangent stiffness tensor is already that precise that the maximum rate of convergence is reached. On the other hand the rate of convergence is very low for *GL-Tjahjanto*. Already 3000 iterations were necessary after half of the simulation time. Employing the logarithmic strain measure, we observe the same number of required iterations for *LOG-Direct* and *LOG-Taxer*. The comparison to a simulation using the analytically correct tangent stiffness tensor revealed that also in this case the maximum rate of convergence is reached. 7% more iterations were required by *LOG-Kalidindi*.

## 7. Conclusions

The methods proposed by the authors are capable of performing a directed strain perturbation much more precisely than other methods proposed in the literature. They are applicable to any material model. Using perturbation magnitudes, $\Delta\varepsilon$ and $\Delta E$, in a range of $10^{-15}$–$10^{-1}$, the discrepancy in generating the correct magnitude of strain perturbation $\Pi^{\text{magnitude}}$ and the discrepancy in conducting the strain perturbation into the correct direction $\Pi^{\text{direction}}$ is smaller by several orders of magnitude compared to the methods proposed by Kalidindi [11] and Miehe [10]. In order to draw a comparison with the approaches presented by Tjahjanto [12] and Taxer [13], we set the perturbation magnitude to $10^{-6}$ and compared the computed tangent stiffness tensor in the case of an isotropic elastic material model with the analytic correct solution. The discrepancy for Tjahjanto's approach is higher by several orders of magnitude than it is for our methods. Taxer achieves slightly better results even though his method and our direct methods, Sections 5.1.1, 5.2.1 and 5.3.1, are just limited by the computational accuracy. The accuracy of Taxer's as well as the authors' method would even increase if the perturbation magnitude is increased to $5 \cdot 10^{-5}$ as it has been proposed in [11]. The investigation of the rate of convergence reveals that depending on the nonlinearity of the material model and the approach for the numerical differentiation, an improved accuracy of the strain perturbation not necessarily increases the rate of convergence. This observation either implies that the maximum rate of convergence is already reached or the error of the numerical differentiation, which was here conducted by applying a forward-difference approximation, might dominate the error arising from the strain perturbation. The correctness of the strain perturbation, however, regains its importance as soon as the perturbation magnitude is reduced or a more sophisticated approximation of the derivative is used (e.g. central-difference or complex-step derivative approximation [17–20]). Once instabilities [22–24] with bifurcation points occur, the tangent stiffness tensor not only defines the rate of convergence, but also the state of equilibrium.

Measuring the system time under identical conditions revealed that the computational effort to determine the tangent stiffness tensor is almost the same for all methods except the one proposed by Taxer, which requires a 8–10 times higher computational effort. Although rigid body rotation is prevented rotational parts can occur for Taxer's method within each perturbation step. Since the Cauchy stress rate is not objective, additional effort is necessary as soon as the utilized material model requires the stress rate. In contrast, our methods guarantee a strain perturbation without any rotational part and therefore do not suffer from a not objective Cauchy stress rate. Applicability of our methods in the context of a more sophisticated plasticity model was demonstrated in a recent contribution [7]. The authors deployed the direct method based on the logarithmic strain measure very successfully within a simulation of a part of an integrated circuit under cyclic thermo-mechanical loading using a crystal plasticity material model.

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